

NURNBERG, GermanyWMO: **107630**Lat: **49.5031N**Lon: **11.0550E**Elev: **318**StdP: **97.57**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.9	-14.9	1.1	-8.9	-11.1	1.5	-6.1	12.3	8.8	10.6	5.8	2.0	10	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.9	19.6	29.8	19.0	27.8	18.3	20.6	28.3	19.8	27.1	19.2	25.8	3.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.6	22.3	17.8	13.2	22.0	16.9	12.5	21.3	60.5	28.8	58.0	27.0	55.4	25.8	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.6	6.4	-13.4	34.9	3.7	1.5	-16.1	35.9	-18.2	36.8	-20.3	37.6	-23.0	38.6		
(5)			-13.8	22.3	3.7	1.0	-16.4	23.0	-18.6	23.6	-20.6	24.2	-23.3	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	0.6	1.0	5.1	10.3	13.8	17.9	19.7	18.6	14.8	10.0	5.2	2.1
	DBStd	7.77	4.45	4.37	4.20	3.98	3.90	3.67	3.42	3.45	3.31	3.80	3.62	4.04
	HDD10.0	1201	293	251	157	44	11	1	0	0	2	48	148	246
	HDD18.3	3223	551	484	411	241	148	52	26	40	115	258	393	504
	CDD10.0	1188	0	0	4	53	128	236	301	266	146	48	5	0
	CDD18.3	171	0	0	0	1	7	38	68	48	9	0	0	0
	CDH23.3	1896	0	0	0	23	109	439	722	501	97	4	0	0
	CDH26.7	622	0	0	0	3	23	138	269	172	17	0	0	0
Wind	WSAvg	2.7	3.1	2.9	3.0	2.6	2.8	2.5	2.6	2.3	2.3	2.5	2.8	3.0
Precipitation	PrecAvg	622	44	35	43	41	63	68	73	64	50	48	47	51
	PrecMax	909	109	112	178	96	151	146	143	175	118	154	125	134
	PrecMin	400	9	5	3	7	12	16	2	3	10	2	1	5
	PrecStd	107	21	22	29	23	31	30	34	31	26	31	21	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.8	19.2	26.0	29.2	33.1	35.0	33.9	29.0	23.9	16.2	12.1
		MCWB	9.6	7.7	10.5	14.3	17.9	20.0	20.5	20.1	18.1	15.9	12.8	9.5
	2%	DB	10.1	10.9	16.9	23.2	26.1	30.1	32.1	31.0	26.1	20.1	14.2	10.2
		MCWB	7.9	7.5	9.7	12.7	16.7	19.0	19.6	19.7	17.5	14.9	11.2	8.4
	5%	DB	8.2	8.9	14.2	21.1	23.9	28.0	29.8	28.2	23.9	18.0	12.2	8.9
		MCWB	6.7	6.4	8.6	12.3	15.4	18.3	18.9	19.0	16.8	14.0	9.9	7.4
	10%	DB	6.8	7.0	12.0	18.9	21.8	25.9	27.2	26.0	21.8	16.0	10.9	7.2
		MCWB	5.7	5.2	8.1	11.5	14.4	17.5	18.2	18.1	16.0	13.1	9.0	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	9.2	11.9	15.4	18.9	21.3	21.2	21.7	19.6	16.9	13.4	10.3
		MCDB	11.9	11.1	16.3	24.0	26.7	29.2	29.4	28.9	25.4	21.2	15.6	11.7
	2%	WB	8.4	7.7	10.6	13.9	17.6	20.0	20.4	20.7	18.4	15.5	11.8	9.0
		MCDB	9.7	10.3	15.1	21.3	24.5	27.4	28.5	28.2	24.1	19.0	13.9	10.3
	5%	WB	6.8	6.4	9.5	12.8	16.3	19.0	19.7	19.7	17.4	14.4	10.3	7.3
		MCDB	8.0	8.5	13.5	19.8	22.3	25.9	27.3	26.3	22.7	17.4	12.3	8.5
	10%	WB	5.3	5.0	8.3	11.9	15.1	18.1	19.1	18.7	16.4	13.4	9.1	6.2
		MCDB	6.3	6.7	11.7	17.9	20.3	24.7	25.8	24.7	20.7	15.9	10.7	7.2
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.4	9.2	11.4	10.7	11.0	11.1	10.9	10.5	8.8	5.9	5.1
		MCDBR	6.8	9.7	14.5	16.0	15.0	15.2	16.1	15.2	14.5	12.0	8.5	6.8
		MCWBR	5.4	6.6	8.2	7.6	6.9	6.1	5.4	5.8	6.7	7.1	6.0	5.3
	5% WB	MCDBR	6.5	8.6	12.3	14.3	13.3	13.4	12.7	13.3	12.2	10.8	7.6	6.1
MCWBR		5.4	6.2	7.5	7.2	6.5	5.7	4.9	5.7	6.1	6.6	5.7	5.1	
Clear-Sky Solar Irradiance	taub	0.306	0.329	0.365	0.398	0.405	0.407	0.416	0.410	0.380	0.356	0.331	0.303	
	taud	2.428	2.372	2.317	2.256	2.267	2.290	2.261	2.301	2.363	2.420	2.449	2.427	
	Ebn at Noon	748	812	839	846	854	853	838	826	814	769	708	696	
	Edh at Noon	63	84	106	124	128	127	128	118	100	80	62	56	
All-Sky Solar Radiation	RadAvg	0.85	1.68	2.77	4.25	5.02	5.56	5.26	4.57	3.33	1.98	0.96	0.67	
	RadStd	0.15	0.29	0.35	0.56	0.57	0.45	0.51	0.43	0.44	0.24	0.12	0.08	